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Dyeing

Created by	Vincent
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Reviewers	Laurent Vincent K Kishore Ganesan

1. Introduction

1.1 Purpose and objectives

This report documents the methodology used to develop new Life Cycle Inventory (LCI) datasets and emission factors for textile dyeing processes, intended for integration into the Carbonfact database. The work aims to replace existing Environmental Footprint (EF) 3.1 datasets, which currently suffer from low transparency, limited methodological documentation, and restricted access to underlying inventory data. This study adopts a transparent, literature-supported, and verifiable LCI approach.

1.2 Functional unit

The functional unit is defined as **1 kg of textile dyeing** (cradle-to-gate), covering core dyeing, washing/rinsing, drying, and padding operations on the specified substrate.

For auxiliary datasets on specialty chemicals and dyes, the functional unit is **1 kg of chemical or dye, cradle-to-gate**, ready to use in textile wet processing.

1.3 System boundaries

The system boundaries cover the following unit processes:

- Core dyeing (dye application)
- Washing and rinsing (post-dye)
- Drying, tumble drying, and padding

The following stages are **excluded** because they are already covered by separate emission factors (EFs) in the Carbonfact database:

- Pre-treatment (covered by `PREPARATION#ANY`)
- Sanforization (covered by `FINISHING/SANFORIZATION`)
- Compact finishing (covered by `FINISHING/COMPACTING/MECHANICAL`)

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2. Data sources and literature review

2.1 Overview of data sources

The following sources were used to determine inputs/outputs and build life cycle inventories of dyeing processes.

📋 **Generic inventory**

- **Supplier primary data** — 20 datasets collected by Carbonfact from suppliers and on-site audits.
- **Emission factor databases** — 15 datasets from EF 3.1, ecoinvent 3.12, and Ecobalyse.

⚡ 💦 **Energy and water inventory**

- **Apparel Impact Institute (Aii) Facility Benchmark Tool** — 214 facility-level datasets on wet processing energy consumption, classified by process group, substrate, fabric, fibre, process description, and dye type.

🧪 **Dyes and auxiliary chemicals inventory**

- **Scientific literature** — 14 datasets from peer-reviewed articles (including Roos et al., Li et al., Yousaf et al., BAT Reference Document for Textiles).
- **Publicly available supplier information** — MSDS sheets and technical datasheets from major textile chemical and dye suppliers (e.g., Archroma).
- **Environmental clearance applications** — Production process data for specific dyes (Disperse Blue 183, Reactive Golden Yellow HER, Pigment Violet 23, Acid Red 87) from Indian regulatory filings.

2.2 Data from Apparel Impact Institute Facility Benchmark

From the Aii benchmark, two process groups were extracted: "Dyeing Only (without Pre-treatment)" (85 datasets) and "Dyeing (full process)" (129 datasets). Energy consumption was included only for steps falling within the system boundaries (dye/print, wash-off, drying/padding). Pre-treatment, preparation, and compacting steps were excluded. For yarn package dyeing, approximately 20% of total energy demand was attributed to pre-processing and was removed from the dyeing allocation.

Raw data from all sources were compiled in a `Data_Sources` database, preserving original naming conventions. The data were then cleaned and harmonised into a standardised `Energy_Database` with consistent column naming and unit conventions (total energy demand in MJ, electricity in kWh, heat in MJ). A classification of data in different dyeing process groups was also performed, as detailed in the next section.

2.3 Key takeaways from literature review

▼ 3. Classification of dyeing processes

Dyeing processes can generally be classified along six parameters:

1. Substrate type (fibre, yarn, fabric, garment)
2. Fabric type (woven, knit, denim)
3. Fibre group (animal fibres, cellulosic fibres, synthetics, blends)
4. Process type (exhaust, continuous, semi-continuous)
5. Process technology (jet, package, CPB, airflow jet, range, etc.)
6. Dye type (disperse, reactive, acid, vat, basic, pigment, indigo).

This classification was built from literature review and benchmarked against ranges provided by the Apparel Impact Institute Facility Benchmark. Two main data validation steps were applied to ensure credibility and refine this classification: a contingency analysis and a parameter sensitivity analysis.

3.1 Contingency analysis

Before running sensitivity tests for each parameter, contingency matrices were built to identify linked parameters (where one parameter fully determines another). Two key linkages were identified:

- **Process technology** ↔ **Process type**: 100% contingency for all named technologies (e.g., jet → exhaust, range → continuous, CPB → semi-continuous).
- **Dye type** ↔ **Fibre group**: strong contingencies (e.g., animal fibres → acid dyes at 100%; cellulosic fibres → reactive at ~80% and vat at ~20%; synthetics → disperse at ~57%, acid at ~30%, basic at ~13%).

Linked parameters were excluded from grouping in the parameter sensitivity analysis below when their counterpart was being tested, to avoid results distortion.

3.2 Parameter sensitivity analysis

Using the `Energy_Database` presented in section 2, a parameter sensitivity analysis was conducted in Python to identify which process classification parameters most significantly influence energy and water consumption.

The approach consisted of sequentially fixing all parameters except one, then measuring the standard deviation (σ) of total energy demand (MJ) and water use (kg) across all values of the varying parameter. For each parameter tested, the analysis produced an average standard deviation across all scenarios and the number of scenarios considered, providing a measure of robustness, as shared in the table below.

PARAMETER	AVERAGE Sdev (σ)	Nb_Scenarios
1. Substrate type	2.7	5
2. Fabric type	2.1	17
3. Fibre group	3.8	15
4. Process type	7.3	17
5. Process technology	7.3	17
6. Dye type	5.2	17

3.3 Key findings and final classification

Fabric type showed the lowest average standard deviation (σ = 2.1 MJ), indicating this parameter does not significantly drive energy consumption variability for a given process where all other 5 parameters are fixed (tested on 5 scenarios).

> ❌ It was therefore excluded from the emission factor modelling rules, and an average EF is computed across fabric types.

Process type and process technology showed the highest average standard deviation (σ = 7.3 MJ), indicating these parameters have a significant influence on energy consumption variability (tested on 17 scenarios each).

> ✅ These 2 parameters were prioritised in the literature review and LCI creation, ensuring coverage of most-common dyeing technologies in the textile industry.

The remaining parameters (substrate type, fibre group and dye type) were also retained as classification axes for the final dataset structure.

▼ 4. Life Cycle Inventory modelling

In this section, all assumptions used to model LCI of dyeing datasets are explored. Extensive documentation can be found in Annex:

- **Annex I:** Full LCI of all dyeing datasets
- **Annex II:** Background dataset source mapping

4.1 Energy consumption

💡 Electricity consumption data were compiled from the harmonised energy database, differentiated by substrate, process technology, and dye type. The baseline electricity dataset used is `market group for electricity, low voltage (GLO)` from `ecoinvent 3.12`.

🔥 Heat consumption values were similarly compiled from the energy database. A custom heat production mix, `Heat, for dyeing process, production mix (GLO)`, was built by Carbonfact using three `ecoinvent 3.12` datasets in the following proportions (derived from the `steam production, as energy carrier, in chemical industry (RoW)` activity):

- 50% — heat production, natural gas, at industrial furnace >100kW (RoW), η = 0.95
- 20% — heat production, at hard coal industrial furnace 1–10MW (RoW), η = 0.80
- 30% — heat production, light fuel oil, at industrial furnace 1MW (RoW), η = 0.95

Key correction applied: The original ecoinvent steam production dataset included a 22% energy loss factor between boiler output and useable process steam. Since the collected LCI data represent actual fuel energy content rather than delivered steam, boiler efficiency was removed by downscaling input quantities using each fuel's efficiency factor (and LHV where necessary).

4.2 Auxiliary chemicals

Specialty chemical inventories were modelled per kg of chemical produced, cradle-to-gate, using `ecoinvent 3.12` background data and literature-based synthesis routes. Climate change impacts (kg CO_2eq/kg chemical) were computed for 24 chemical function categories (wetting agents, scouring agents, electrolytes, dispersing agents, levelling agents, fixing agents, pH control agents, softeners, etc.).

Chemical dosing rules differ by process type:

- **Exhaust dyeing**: chemical recipes are expressed in g/L. A default liquor ratio is assumed per substrate/technology/dye type combination, determining absolute chemical quantities per kg of textile.
- **Continuous and CPB dyeing**: chemical recipes are expressed in % on weight of fabric (OWF). The pick-up rate determines the actual chemical load.

A fixed repartition of the chemical inventory is applied within each process category, while the absolute quantity scales with the liquor ratio or pick-up rate.

4.3 Dyestuff production

Since detailed dye composition and production data are not publicly available, dye LCIs were developed using a bottom-up, industry-informed approach. For each dye class, a representative production route was identified from environmental clearance applications and literature, and the inventory was constructed using ecoinvent proxy datasets for intermediates with comparable chemistry. Four dye production LCIs were built:

- **Disperse dye** (azo, diazotisation–coupling route, based on Disperse Blue 183): total GHG impact = 6.53 kg CO_2e/kg dye. Dominant contributors: 2-nitroaniline (41%), spray-drying steam (30%).
- **Reactive dye** (azo, monochlorotriazine route, based on Reactive Golden Yellow HER): total GHG impact = 6.37 kg CO_2e/kg dye. Dominant contributor: spray-drying steam (42%), with multiple intermediates each contributing 7–14%.
- **Vat dye** (anthraquinone, alkali fusion route, based on Pigment Violet 23): total GHG impact = 7.21 kg CO_2e/kg dye. Dominant contributors: glycine/N-phenylglycine proxy (64%), steam (22%).
- **Acid dye** (bromination route, based on Acid Red 87): total GHG impact = 8.27 kg CO_2e/kg dye. Dominant contributors: 2-nitroaniline (52%), bromine (19%), steam (15%).

Indigo pigments for denim rope dyeing were not modelled from primary data; anthraquinone-based vat dyes were used as proxy.

4.4 Water consumption

Water consumption was compiled per dyeing route from BAT reference documents, literature, supplier data, and extrapolations based on liquor ratio. Values range from ~17 L/kg (loose fibre acid dyeing) to ~167 L/kg (conventional jet, acid, animal fibres). Continuous processes (CPB, pad-steam, thermosol) show the lowest water consumption (~20–36 L/kg), while exhaust processes in conventional jet machines show the highest (~100–167 L/kg).

4.5 Wastewater

Wastewater was estimated assuming 10% of process water is lost through evaporation during heating and drying, with the remaining 90% discharged as effluent. At this stage, the generic `market for wastewater, average` dataset from ecoinvent 3.12 was used, as the current assessment prioritises climate change impact and variations in pollutant load have limited influence on GHG emissions. More detailed textile-specific wastewater datasets will be needed for future multi-impact assessments (eutrophication, ecotoxicity, water scarcity).

4.6 Emission factors used

▼ 5. Life Cycle Impact Assessment

5.1 LCIA method and scope

Impacts were computed using Brightway2 for climate change (kg CO_2eq). Energy use consistently dominates the carbon footprint, typically contributing 80–95% of total emissions across all routes.

5.2 Results by dyeing route

Dyeing route	GHG impact (kgCO2e/kg)
Fiber dyeing, Exhaust, Package, Acid, Synthetic Fibers, Medium Shade	3.58
Yarn dyeing, Exhaust, Package, Acid, Animal Fibers, Medium Shade	4.18
Fabric dyeing, Continuous, Pad steam, Acid, Animal Fibers, Medium Shade	3.29
Fabric dyeing, Continuous, Thermosol, Disperse, Synthetic Fibers, Medium Shade	3.09
Fabric dyeing, Exhaust, Softflow jet, Acid, Synthetic Fibers, Medium Shade	3.18
Fabric dyeing, Exhaust, Conventional jet, Disperse, Synthetic Fibers, Medium Shade	4.76
Fabric dyeing, Exhaust, Softflow jet, Disperse, Synthetic Fibers, Medium Shade	3.44
Fabric dyeing, Exhaust, Softflow jet, Acid, Animal Fibers, Medium Shade	4.68
Yarn dyeing, Exhaust, Hank, Disperse, Synthetic Fibers, Medium Shade	4.85
Fabric dyeing, Exhaust, Beam, Disperse, Synthetic Fibers, Medium Shade	4.08
Fabric dyeing, Exhaust, Beam, Acid, Synthetic Fibers, Medium Shade	3.21
Yarn dyeing, Exhaust, Package, Disperse, Synthetic Fibers, Medium Shade	3.82
Fabric dyeing, Exhaust, Softflow jet, Reactive, Cellulosic Fibers, Medium Shade	3.98
Yarn dyeing, Exhaust, Package, Acid, Synthetic Fibers, Medium Shade	3.67
Fabric dyeing, Exhaust, Beam, Reactive, Cellulosic Fibers, Medium Shade	4.07
Fabric dyeing, Exhaust, Beam, Acid, Animal Fibers, Medium Shade	5.00
Fiber dyeing, Exhaust, Package, Vat, Cellulosic Fibers, Medium Shade	4.01
Yarn dyeing, Exhaust, Package, Vat, Cellulosic Fibers, Medium Shade	5.11
Yarn dyeing, Exhaust, Hank, Vat, Cellulosic Fibers, Medium Shade	8.48
Fabric dyeing, Exhaust, Airflow jet, Disperse, Synthetic Fibers, Medium Shade	2.34
Fabric dyeing, Exhaust, Conventional jet, Acid, Animal Fibers, Medium Shade	6.79
Fiber dyeing, Exhaust, Package, Reactive, Cellulosic Fibers, Medium Shade	3.97
Yarn dyeing, Exhaust, Hank, Reactive, Cellulosic Fibers, Medium Shade	5.75
Fabric dyeing, Exhaust, Airflow jet, Acid, Animal Fibers, Medium Shade	3.26
Fabric dyeing, Continuous, Pad steam, Vat, Cellulosic Fibers, Medium Shade	3.63
Fiber dyeing, Exhaust, Package, Acid, Animal Fibers, Medium Shade	4.14
	2.04

Dyeing route	GHG impact (kgCO2e/kg)
Fabric dyeing, Exhaust, Airflow jet, Acid, Synthetic Fibers, Medium Shade	
Yarn dyeing, Exhaust, Package, Reactive, Cellulosic Fibers, Medium Shade	5.19
Fabric dyeing, Exhaust, Airflow jet, Reactive, Cellulosic Fibers, Medium Shade	2.49
Fabric dyeing, Exhaust, Conventional jet, Reactive, Cellulosic Fibers, Medium Shade	5.54
Fabric dyeing, Exhaust, Conventional jet, Acid, Synthetic Fibers, Medium Shade	4.33
Fiber dyeing, Exhaust, Package, Disperse, Synthetic Fibers, Medium Shade	4.19
Yarn dyeing, Exhaust, Hank, Acid, Synthetic Fibers, Medium Shade	3.98
Fabric dyeing, Continuous, Pad steam, Reactive, Cellulosic Fibers, Medium Shade	3.39
Garment dyeing, Exhaust, Front Loader, Reactive, Cellulosic Fibers, Medium Shade	4.45
Yarn dyeing, Continuous, Rope, Indigo, Cellulosic Fibers, Medium Shade	7.89
Yarn dyeing, Exhaust, Hank, Acid, Animal Fibers, Medium Shade	6.72
Fabric dyeing, Semi-continuous, CPB, Reactive, Cellulosic Fibers, Medium Shade	2.81

Key observations:

- **Continuous processes** (thermosol, pad-steam, CPB) show the lowest impacts (~2.9–3.8 kg CO_2e/kg), owing to low liquor ratios and short residence times.
- **Airflow jet exhaust** processes also perform well (~2.1–3.4 kg CO_2e/kg) due to air-assisted fabric transport reducing liquor requirements.
- **Conventional jet** and **beam dyeing** exhaust processes produce higher emissions (~4.3–7.0 kg CO_2e/kg), driven by large bath volumes and repeated heating.
- **Softflow jet** processes show moderate total emissions (~3.3–4.9 kg CO_2e/kg) but with a notably higher share of chemical contributions (up to ~41%) versus energy.

5.3 Comparison with EF 3.1 datasets

The comparison with EF 3.1 highlights a fundamental limitation: EF datasets aggregate multiple machine types into broad "batch" or "continuous" categories without differentiating specific technologies (airflow vs. conventional jet, pad-steam vs. thermosol vs. CPB). Important drivers—particularly liquor ratio—are averaged out. Differences between the Carbonfact results and EF range from −52% to +86%, depending on the route. The Carbonfact model captures technology-specific variation that the EF production-mix approach masks.

5.4 Comparison with Roos et al.

Total energy demand in both models is broadly comparable (~30–35 MJ per kg fabric), confirming similar underlying process physics. Energy contribution to total carbon footprint is also similar (70–90%). The divergence in absolute footprint values (Carbonfact: ~4–7 kg CO_2e vs. Roos: ~1.5–2.0 kg CO_2e) is primarily driven by differences in energy carrier mix and associated emission factors, not by process energy quantities.

▼ 6. Sensitivity analysis

6.1 Electricity source sensitivity

Using reactive exhaust dyeing on conventional jet as the test case (highest-impact route), substitution of the global electricity mix with alternative sources shows a total footprint variation of −19% (solar PV) to +9% (Bangladesh grid), corresponding to 5.33–7.22 kg CO_2e/kg fabric.

6.2 Thermal energy source sensitivity

Thermal energy substitution has a stronger influence: switching from the baseline industrial steam mix (RoW) to natural gas boilers reduces emissions by ~17–33%, while coal-based heat increases them by up to +13%. Biomass co-generation (wood chips) achieves the largest reduction (~46%), bringing the total to ~3.55 kg CO_2e/kg fabric.

6.3 Conclusion on sensitivity

Energy sources—particularly thermal energy—are the dominant lever for reducing the carbon footprint of textile dyeing. The choice of thermal energy source alone can alter the total footprint by approximately ±45%.

▼ 7. Dataset integration into Carbonfact

7.1 Dataset descriptor hierarchy

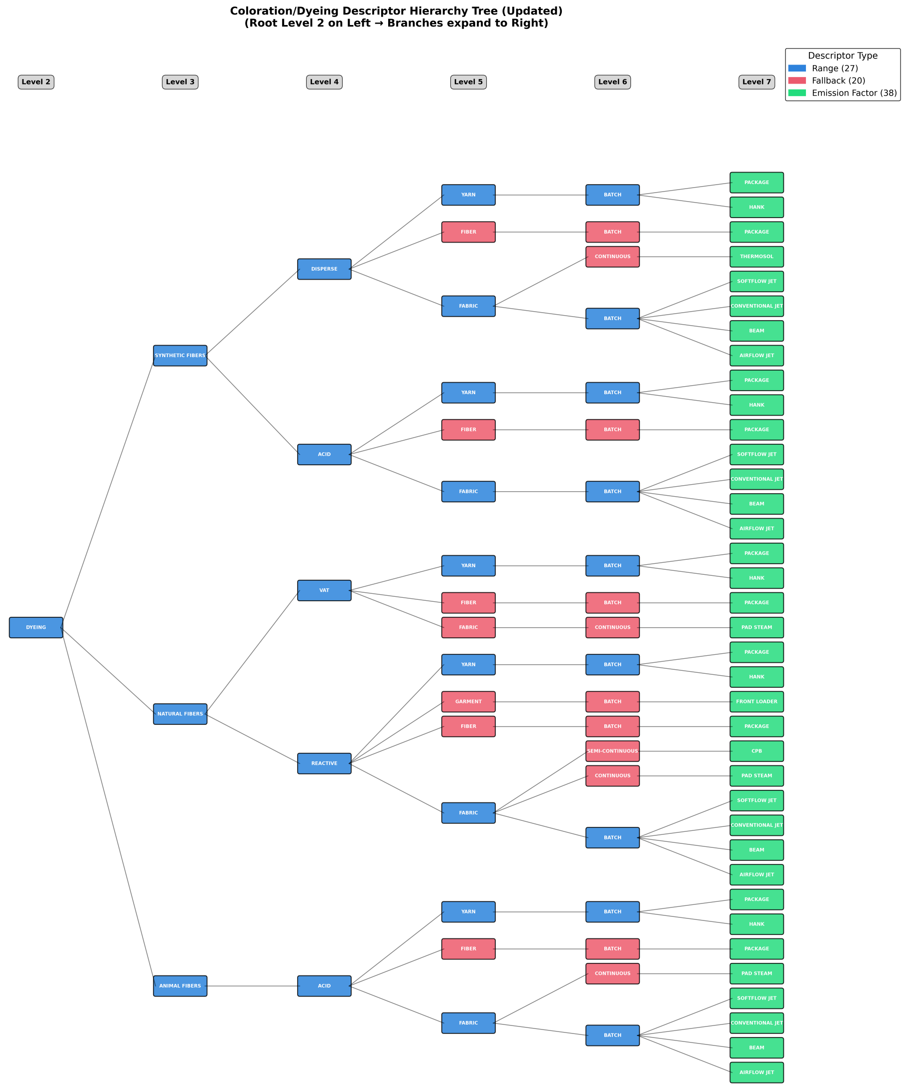
Datasets are structured using a 7-level hierarchy representing key process parameters:

1. Process category 1 (e.g., COLORATION)
2. Process category 2 (e.g., DYEING)
3. Fibre group (e.g., SYNTHETIC_FIBERS)
4. Dye type (e.g., DISPERSE)
5. Substrate type (e.g., FABRIC)
6. Process type (e.g., BATCH)
7. Process technology (e.g., AIRFLOW_JET)

Full descriptor example: `COLORATION/DYEING/SYNTHETIC_FIBERS/DISPERSE/FABRIC/BATCH/AIRFLOW_JET`

7.2 Dataset construction logic

All level-7 datasets were calculated using custom LCIs computed with Brightway. Lower-level datasets (levels 1–6) are derived either as ranges (min/median/max across child nodes) or as fallbacks (when only one child node exists). This tree logic allows the platform to serve the most specific dataset available given user inputs, while defaulting to aggregated values when detailed parameters are unknown.



7.3 Disaggregated energy structure

A disaggregated energy structure is embedded in each dataset's JSON, reporting the share and absolute quantity of electricity (kWh) and heat (MJ) separately. This enables: energy regionalisation at scale (applying country-specific emission factors), supplier-level energy specification, and granular decarbonisation modelling.

▼ 8. Limitations and future work

8.1 Current limitations

- Dye production LCIs rely on proxy intermediates from ecoinvent due to proprietary formulations, introducing moderate uncertainty.
- Wastewater is modelled generically; textile-specific pollutant loads are not yet reflected.
- Softflow jet data would benefit from additional primary data validation.
- Indigo pigments for denim use an anthraquinone vat dye proxy rather than a dedicated LCI.
- Several water consumption values are extrapolated from liquor ratios rather than measured directly.

- The sensitivity analysis was conducted on a single dyeing route; extending it to other routes would strengthen conclusions.

8.2 Future improvements

- Develop textile-specific wastewater treatment datasets for multi-impact assessments (eutrophication, ecotoxicity, water scarcity).
- Incorporate primary data from additional suppliers and facility audits to reduce reliance on literature averages.
- Model shade depth influence on chemical and dye consumption.
- Expand coverage to include solution/dope dyeing, waterless/supercritical CO_2 dyeing, and biological dyeing routes currently listed in the EFS but not yet modelled.
- Integrate region-specific energy mixes at the dataset level to support country-level carbon footprinting.

▼ 9. References

- Apparel Impact Institute, Facility Benchmark Calculator (v0.9, 2025)
- BAT Reference Document for Textiles (European Commission)
- ecoinvent database v3.12 (Cut-off system model)
- Environmental Footprint (EF) 3.1 datasets
- Ecobalyse database
- Li X, Zhu L, Ding X, Wu X, Wang L. Climate Change and the Textile Industry: The Carbon Footprint of Dyes. *AATCC Journal of Research*. 2023;11(2):109-123.
- Roos S et al. — referenced for dyeing process energy and carbon footprint benchmarking
- Yousaf et al. 2023 — textile dye classification
- Environmental clearance applications: Parishi Chemicals (Disperse Blue 183), Aditayam Industries (Reactive Golden Yellow HER), Agrasem Colours (Pigment Violet 23), Agrasem Industries (Acid Red 87)
- Detailed specialty chemicals and LCI data: Google Sheets workbook

▼ 10. Annexes

ℹ️ Annex I: Full LCI of all dyeing datasets

Activity name	Reference product	Location	Unit	Dyeing classification	Process type	Process technology	Dyei
Fabric dyeing, Exhaust, Airflow jet, Acid, Animal Fibers, Medium Shade	dyed fabric, airflow jet, acid, animal fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Airflow jet	Acid
Fabric dyeing, Exhaust, Airflow jet, Acid, Synthetic Fibers, Medium Shade	dyed fabric, airflow jet, acid, synthetic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Airflow jet	Acid

Fabric dyeing, Exhaust, Airflow jet, Disperse, Synthetic Fibers, Medium Shade	dyed fabric, airflow jet, disperse, synthetic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Airflow jet	Dispe
Fabric dyeing, Exhaust, Airflow jet, Reactive, Cellulosic Fibers, Medium Shade	dyed fabric, airflow jet, reactive, cellulosic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Airflow jet	Reac
Fabric dyeing, Exhaust, Softflow jet, Acid, Animal Fibers, Medium Shade	dyed fabric, softflow jet, acid, animal fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Softflow jet	Acid
Fabric dyeing, Exhaust, Softflow jet, Acid, Synthetic Fibers, Medium Shade	dyed fabric, softflow jet, acid, synthetic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Softflow jet	Acid
Fabric dyeing, Exhaust, Softflow jet, Disperse, Synthetic Fibers, Medium Shade	dyed fabric, softflow jet, disperse, synthetic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Softflow jet	Dispe
Fabric dyeing, Exhaust, Softflow jet, Reactive, Cellulosic Fibers, Medium Shade	dyed fabric, softflow jet, reactive, cellulosic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Softflow jet	Reac
Fabric dyeing, Exhaust, Conventional jet, Acid, Animal Fibers, Medium Shade	dyed fabric, conventional jet, acid, animal fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Conventional jet	Acid
Fabric dyeing, Exhaust, Conventional jet, Acid, Synthetic Fibers, Medium Shade	dyed fabric, conventional jet, acid, synthetic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Conventional jet	Acid
Fabric dyeing, Exhaust, Conventional jet, Disperse, Synthetic Fibers, Medium Shade	dyed fabric, conventional jet, disperse, synthetic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Conventional jet	Dispe
Fabric dyeing, Exhaust,	dyed fabric, conventional	GLO	kg	Fabric dyeing	Exhaust	Conventional jet	Reac

Conventional jet, Reactive, Cellulosic Fibers, Medium Shade	jet, reactive, cellulosic fibers, medium shade						
Fabric dyeing, Exhaust, Beam, Acid, Animal Fibers, Medium Shade	dyed fabric, beam, acid, animal fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Beam	Acid
Fabric dyeing, Exhaust, Beam, Acid, Synthetic Fibers, Medium Shade	dyed fabric, beam, acid, synthetic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Beam	Acid
Fabric dyeing, Exhaust, Beam, Disperse, Synthetic Fibers, Medium Shade	dyed fabric, beam, disperse, synthetic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Beam	Dispe
Fabric dyeing, Exhaust, Beam, Reactive, Cellulosic Fibers, Medium Shade	dyed fabric, beam, reactive, cellulosic fibers, medium shade	GLO	kg	Fabric dyeing	Exhaust	Beam	Reac
Fabric dyeing, Semi-continuous, CPB, Reactive, Cellulosic Fibers, Medium Shade	dyed fabric, cpb, reactive, cellulosic fibers, medium shade	GLO	kg	Fabric dyeing	Semi-continuous	CPB	Reac
Fabric dyeing, Continuous, Pad steam, Acid, Animal Fibers, Medium Shade	dyed fabric, pad steam, acid, animal fibers, medium shade	GLO	kg	Fabric dyeing	Continuous	Pad steam	Acid
Fabric dyeing, Continuous, Pad steam, Reactive, Cellulosic Fibers, Medium Shade	dyed fabric, pad steam, reactive, cellulosic fibers, medium shade	GLO	kg	Fabric dyeing	Continuous	Pad steam	Reac
Fabric dyeing, Continuous, Pad steam, Vat, Cellulosic Fibers, Medium Shade	dyed fabric, pad steam, vat, cellulosic fibers, medium shade	GLO	kg	Fabric dyeing	Continuous	Pad steam	Vat
Fabric dyeing, Continuous, Thermosol, Disperse, Synthetic Fibers, Medium Shade	dyed fabric, thermosol, disperse, synthetic fibers, medium shade	GLO	kg	Fabric dyeing	Continuous	Thermosol	Dispe
Yarn dyeing, Exhaust,	dyed yarn, package, acid,	GLO	kg	Yarn dyeing	Exhaust	Package	Acid

Package, Acid, Animal Fibers, Medium Shade	animal fibers, medium shade						
Yarn dyeing, Exhaust, Package, Acid, Synthetic Fibers, Medium Shade	dyed yarn, package, acid, synthetic fibers, medium shade	GLO	kg	Yarn dyeing	Exhaust	Package	Acid
Yarn dyeing, Exhaust, Package, Disperse, Synthetic Fibers, Medium Shade	dyed yarn, package, disperse, synthetic fibers, medium shade	GLO	kg	Yarn dyeing	Exhaust	Package	Dispe
Yarn dyeing, Exhaust, Package, Reactive, Cellulosic Fibers, Medium Shade	dyed yarn, package, reactive, cellulosic fibers, medium shade	GLO	kg	Yarn dyeing	Exhaust	Package	Reac
Yarn dyeing, Exhaust, Package, Vat, Cellulosic Fibers, Medium Shade	dyed yarn, package, vat, cellulosic fibers, medium shade	GLO	kg	Yarn dyeing	Exhaust	Package	Vat
Yarn dyeing, Exhaust, Hank, Acid, Animal Fibers, Medium Shade	dyed yarn, hank, acid, animal fibers, medium shade	GLO	kg	Yarn dyeing	Exhaust	Hank	Acid
Yarn dyeing, Exhaust, Hank, Acid, Synthetic Fibers, Medium Shade	dyed yarn, hank, acid, synthetic fibers, medium shade	GLO	kg	Yarn dyeing	Exhaust	Hank	Acid
Yarn dyeing, Exhaust, Hank, Disperse, Synthetic Fibers, Medium Shade	dyed yarn, hank, disperse, synthetic fibers, medium shade	GLO	kg	Yarn dyeing	Exhaust	Hank	Dispe
Yarn dyeing, Exhaust, Hank, Reactive, Cellulosic Fibers, Medium Shade	dyed yarn, hank, reactive, cellulosic fibers, medium shade	GLO	kg	Yarn dyeing	Exhaust	Hank	Reac
Yarn dyeing, Exhaust, Hank, Vat, Cellulosic Fibers, Medium Shade	dyed yarn, hank, vat, cellulosic fibers, medium shade	GLO	kg	Yarn dyeing	Exhaust	Hank	Vat
Yarn dyeing, Continuous, Rope, Indigo, Cellulosic Fibers, Medium Shade	dyed yarn, rope, indigo, cellulosic fibers, medium shade	GLO	kg	Yarn dyeing	Continuous	Rope	Indig

Garment dyeing, Exhaust, Front Loader, Reactive, Cellulosic Fibers, Medium Shade	dyed garment, front loader, reactive, cellulosic fibers, medium shade	GLO	kg	Garment dyeing	Exhaust	Front Loader	Reac
Fiber dyeing, Exhaust, Package, Acid, Animal Fibers, Medium Shade	dyed garment, package, acid, animal fibers, medium shade	GLO	kg	Fiber dyeing	Exhaust	Package	Acid
Fiber dyeing, Exhaust, Package, Acid, Synthetic Fibers, Medium Shade	dyed garment, package, acid, synthetic fibers, medium shade	GLO	kg	Fiber dyeing	Exhaust	Package	Acid
Fiber dyeing, Exhaust, Package, Disperse, Synthetic Fibers, Medium Shade	dyed garment, package, disperse, synthetic fibers, medium shade	GLO	kg	Fiber dyeing	Exhaust	Package	Dispe
Fiber dyeing, Exhaust, Package, Reactive, Cellulosic Fibers, Medium Shade	dyed garment, package, reactive, cellulosic fibers, medium shade	GLO	kg	Fiber dyeing	Exhaust	Package	Reac
Fiber dyeing, Exhaust, Package, Vat, Cellulosic Fibers, Medium Shade	dyed garment, package, vat, cellulosic fibers, medium shade	GLO	kg	Fiber dyeing	Exhaust	Package	Vat

ℹ️ Annex II: Background dataset source mapping

Input/Output name	Exchange name	Reference product	Location	Type	Database
Acetic acid	market for acetic acid	acetic acid	GLO	technosphere	ecoinvent-3.12-cut
Acid dyes	Azo acid dyes, from bromisation production	azo acid dyes, from bromisation	GLO	technosphere	Carbonfact_Dyeing
Ammonium sulfate	market for ammonium sulfate	ammonium sulfate	RoW	technosphere	ecoinvent-3.12-cut
Anti migrating agent	market for polyacrylamide	polyacrylamide	GLO	technosphere	ecoinvent-3.12-cut
Caustic soda	market for sodium hydroxide, without water, in 50% solution state	sodium hydroxide, without water, in 50% solution state	RoW	technosphere	ecoinvent-3.12-cut
Disperse dyes	Azo disperse dyes, from diazotization-coupling production	azo disperse dyes, from diazotization-coupling	GLO	technosphere	Carbonfact_Dyeing

Dispersing agent	market for acrylic dispersion, with water, in 58% solution state	acrylic dispersion, with water, in 58% solution state	RoW	technosphere	ecoinvent-3.12-cut
Emulsion	market for silicone product	silicone product	RoW	technosphere	ecoinvent-3.12-cut
Fixing agent	market for acetic acid	acetic acid	GLO	technosphere	ecoinvent-3.12-cut
Formic acid	market for formic acid	formic acid	RoW	technosphere	ecoinvent-3.12-cut
Glauber salt	market for sodium sulfate, anhydrite	sodium sulfate, anhydrite	RoW	technosphere	ecoinvent-3.12-cut
Grid mix electricity	market group for electricity, low voltage	electricity, low voltage	GLO	technosphere	ecoinvent-3.12-cut
Heat mix	Heat, for dyeing process, production mix	heat, for dyeing process	GLO	technosphere	Carbonfact_Dyeing
Indigo	Anthraquinone vat dyes, from alkali fusion production	anthraquinone vat dyes, from alkali fusion	GLO	technosphere	Carbonfact_Dyeing
Levelling agent	market for non-ionic surfactant	non-ionic surfactant	GLO	technosphere	ecoinvent-3.12-cut
Lubricant	market for organophosphorus-compound, unspecified	organophosphorus-compound, unspecified	GLO	technosphere	ecoinvent-3.12-cut
Non-ionic surfactant	market for non-ionic surfactant	non-ionic surfactant	GLO	technosphere	ecoinvent-3.12-cut
Reactive dyes	Azo reactive dyes, from monochlorotriazine production	azo reactive dyes, from monochlorotriazine	GLO	technosphere	Carbonfact_Dyeing
Reduction clearing agent	market for sodium dithionite, anhydrous	sodium dithionite, anhydrous	RoW	technosphere	ecoinvent-3.12-cut
Sequestering agent	market for EDTA, ethylenediaminetetraacetic acid	EDTA, ethylenediaminetetraacetic acid	GLO	technosphere	ecoinvent-3.12-cut
Soaping agent	market for non-ionic surfactant	non-ionic surfactant	GLO	technosphere	ecoinvent-3.12-cut
Soda ash	market for soda ash, light	soda ash, light	RoW	technosphere	ecoinvent-3.12-cut
Sodium carbonate	market for soda ash, light	soda ash, light	RoW	technosphere	ecoinvent-3.12-cut
Sodium hydrosulphite	market for sodium hydrosulfide	sodium hydrosulfide	GLO	technosphere	ecoinvent-3.12-cut
Sodium hydroxide	market for sodium hydroxide, without water, in 50% solution state	sodium hydroxide, without water, in 50% solution state	RoW	technosphere	ecoinvent-3.12-cut
Softener	market for ethoxylated alcohol (AE>20)	ethoxylated alcohol (AE>20)	GLO	technosphere	ecoinvent-3.12-cut
Stabilizer	market for magnesium sulfate	magnesium sulfate	GLO	technosphere	ecoinvent-3.12-cut
Tap water	market group for tap water	tap water	GLO	technosphere	ecoinvent-3.12-cut
Urea	market for urea	urea	RoW	technosphere	ecoinvent-3.12-cut
Vat	Anthraquinone vat dyes, from alkali fusion production	anthraquinone vat dyes, from alkali fusion	GLO	technosphere	Carbonfact_Dyeing
Wastewater, textile industry	market for wastewater from textile production	wastewater from textile production	RoW	technosphere	ecoinvent-3.12-cut
Water to air	Water		GLO	biosphere	ecoinvent-3.12-biosphere
Wetting agent	market for non-ionic surfactant	non-ionic surfactant	GLO	technosphere	ecoinvent-3.12-cut

Activity	GHG (kgCO2e/kg)
Fiber dyeing, Exhaust, Package, Acid, Synthetic Fibers, Medium Shade	3.582364807147280
Yarn dyeing, Exhaust, Package, Acid, Animal Fibers, Medium Shade	4.184468675512870
Fabric dyeing, Continuous, Pad steam, Acid, Animal Fibers, Medium Shade	3.2935744511858500
Fabric dyeing, Continuous, Thermosol, Disperse, Synthetic Fibers, Medium Shade	3.089605055077760
Fabric dyeing, Exhaust, Softflow jet, Acid, Synthetic Fibers, Medium Shade	3.175572903784040
Fabric dyeing, Exhaust, Conventional jet, Disperse, Synthetic Fibers, Medium Shade	4.758447469054420
Fabric dyeing, Exhaust, Softflow jet, Disperse, Synthetic Fibers, Medium Shade	3.438927206697010
Fabric dyeing, Exhaust, Softflow jet, Acid, Animal Fibers, Medium Shade	4.676269107367370
Yarn dyeing, Exhaust, Hank, Disperse, Synthetic Fibers, Medium Shade	4.848481340130330
Fabric dyeing, Exhaust, Beam, Disperse, Synthetic Fibers, Medium Shade	4.077680985799850
Fabric dyeing, Exhaust, Beam, Acid, Synthetic Fibers, Medium Shade	3.2134808720202600
Yarn dyeing, Exhaust, Package, Disperse, Synthetic Fibers, Medium Shade	3.8224904730335400
Fabric dyeing, Exhaust, Softflow jet, Reactive, Cellulosic Fibers, Medium Shade	3.9754449440006100
Yarn dyeing, Exhaust, Package, Acid, Synthetic Fibers, Medium Shade	3.669416846450620
Fabric dyeing, Exhaust, Beam, Reactive, Cellulosic Fibers, Medium Shade	4.073651786834030
Fabric dyeing, Exhaust, Beam, Acid, Animal Fibers, Medium Shade	4.996023459085460
Fiber dyeing, Exhaust, Package, Vat, Cellulosic Fibers, Medium Shade	4.0090903916444500
Yarn dyeing, Exhaust, Package, Vat, Cellulosic Fibers, Medium Shade	5.1096169554782600
Yarn dyeing, Exhaust, Hank, Vat, Cellulosic Fibers, Medium Shade	8.475278166333080
Fabric dyeing, Exhaust, Airflow jet, Disperse, Synthetic Fibers, Medium Shade	2.337166494137970
Fabric dyeing, Exhaust, Conventional jet, Acid, Animal Fibers, Medium Shade	6.790976691216100
Fiber dyeing, Exhaust, Package, Reactive, Cellulosic Fibers, Medium Shade	3.965713118689250
Yarn dyeing, Exhaust, Hank, Reactive, Cellulosic Fibers, Medium Shade	5.748453212171090
Fabric dyeing, Exhaust, Airflow jet, Acid, Animal Fibers, Medium Shade	3.256891377378600
Fabric dyeing, Continuous, Pad steam, Vat, Cellulosic Fibers, Medium Shade	3.6257725365316800
Fiber dyeing, Exhaust, Package, Acid, Animal Fibers, Medium Shade	4.141622654716960
Fabric dyeing, Exhaust, Airflow jet, Acid, Synthetic Fibers, Medium Shade	2.038228675280970

Yarn dyeing, Exhaust, Package, Reactive, Cellulosic Fibers, Medium Shade	5.187069974839000
Fabric dyeing, Exhaust, Airflow jet, Reactive, Cellulosic Fibers, Medium Shade	2.4917508711835200
Fabric dyeing, Exhaust, Conventional jet, Reactive, Cellulosic Fibers, Medium Shade	5.542709522974190
Fabric dyeing, Exhaust, Conventional jet, Acid, Synthetic Fibers, Medium Shade	4.334459661886250
Fiber dyeing, Exhaust, Package, Disperse, Synthetic Fibers, Medium Shade	4.190749916076740
Yarn dyeing, Exhaust, Hank, Acid, Synthetic Fibers, Medium Shade	3.980108970434220
Fabric dyeing, Continuous, Pad steam, Reactive, Cellulosic Fibers, Medium Shade	3.391850171030040
Garment dyeing, Exhaust, Front Loader, Reactive, Cellulosic Fibers, Medium Shade	4.4496205781669400
Yarn dyeing, Continuous, Rope, Indigo, Cellulosic Fibers, Medium Shade	7.894678985932880
Yarn dyeing, Exhaust, Hank, Acid, Animal Fibers, Medium Shade	6.720699389493400
Fabric dyeing, Semi-continuous, CPB, Reactive, Cellulosic Fibers, Medium Shade	2.8123590719908200